Luis Mastrangelo

Software Development Engineer

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Education

Apr 2013–June 2019 **Ph.D. Graduate, Computer Science**, *University of Lugano*, Switzerland.
Programming languages research in Java, Scala and Haskell, runtime systems and virtual machines with C++, software verification, data analysis using R and Python.

Apr 2007–Nov 2011 **Licentiate in Computer Science**, *University of Buenos Aires*, Argentina.
This degree is considered by many institutions as being equivalent to an M.Sc. degree.

Apr 2011–Oct 2011 **Internship**, *INRIA/University of Strasbourg*, France.
Internship title: "A Virtual Machine for Automatic Program Parallelization".

Apr 2003–Dec 2006 **Computer Analyst**, *University of Buenos Aires*, Argentina.
This is a 3 years degree as part of the *Licentiate in Computer Science* programme.

Professional Experience

Nov 2023–Mar 2026 **Senior Software Engineer**, *Hashgraph (ex. Swirlds Labs)*, Remote.
Smart Contracts team, tooling to support EVM development in Hedera networks

Apr 2023–May 2023 **Senior Software Engineer**, *ChainSafe*, Remote.
Worked in Lodestar, an Ethereum Consensus Client written in TypeScript

Jul 2022–Jan 2023 **Senior Blockchain Engineer**, *ConsenSys*, Remote.
Truffle team, a development environment for Ethereum Solidity Smart Contracts

May 2022–Jun 2022 **Lead Blockchain Engineer**, *Campground*, Remote.
NEAR Smart Contract development with Rust and TypeScript

Sep 2020–Apr 2022 **Senior Software Engineer**, *EPAM Systems*, Zurich, Switzerland.
Ethereum & NEAR blockchains development with Solidity, Rust and TypeScript

Aug 2019–Jan 2020 **Technical Consultant**, *DB&L Services SA for Avaloq*, Lugano, Switzerland.
IT infrastructure support for Monitoring Operations at Avaloq Sourcing (CH & LI) SA.

Dec 2006–Mar 2011 **Project Leader**, *Recursiva S.R.L.*, Buenos Aires, Argentina.
Software engineering and architect roles using Java and C# among others technologies.

Mar–Jul 2006 **Network Administrator**, *HS Eventos*, Buenos Aires, Argentina.
Technical service, software installation and maintenance of company network.

Jul–Oct 2004 **Network Administrator**, *Texpampa*, Buenos Aires, Argentina.
Technical service, software installation and maintenance of company network.

Mar 2000–May 2004 **Software Developer**, *Fernando Hermo y Asoc.*, Buenos Aires, Argentina.
ERP and CRM development using VB 6, SQL Server and Crystal Reports.

Freelance & Open Source Projects

Apr 2023–Mar 2025 **Software Engineer**, *SEVM Open Source project*, Remote.
A Symbolic EVM bytecode interpreter, parser and decompiler. Written in TypeScript.

Jan–May 2020 **Software Engineer**, *GTM Dashboard Open Source project*, Remote.
Web dashboard for Git Time Tracking. Written in Go, Rust, TypeScript and Svelte.

Feb 2014–Jan 2020 **Software Developer**, *JNIF*, Lugano, Switzerland.
Java Native Instrumentation Framework written in C++.

May–Sep 2018 **Software Developer**, *Essentia, Family Office*, Lugano, Switzerland.
Desktop application written in Scala.

May–Jun 2018 **Software Engineer**, *GTM Open Source project (contribution)*, Remote.
Add support for submodules in Git Time Tracking. Written in Go. Pull Request.

Oct–Nov 2017 **Software Developer**, *Essentia, Family Office*, Lugano, Switzerland.
Desktop application written in C#.

Apr 2016–Oct 2017 **Software Developer**, *HSC*, Remote.
A Supercompiler prototype for Haskell.

Dec 2010–Jan 2011 **Web Developer**, *vCelebrate*, Buenos Aires, Argentina.
Web application using Facebook API and Adobe Flex.

Technical Skills

Languages	Solidity, Blockchain, Ethereum, NEAR, Smart Contracts, Rust, Java, Kotlin, Scala, JavaScript/TypeScript, C/C++, C#, Haskell, Go, Python, R, SQL, UML
Paradigms & Fields	OOP, Function Programming, Static & Dynamic Analysis, Data Science
Databases	Design, Oracle PL/SQL, Datalog/QL, PostgreSQL, MySQL, SQL Server, SQLite
Editors & Tools	VSCode, Git, IntelliJ/CLion/Rider, Eclipse, Xcode, Vim/Emacs, Linux, Apache

Teaching Experience

Apr 2013–Nov 2017 **Teaching Assistant**, *University of Lugano*, Switzerland. Multiple courses.

Apr 2006–Nov 2010 **Teaching Assistant**, *University of Buenos Aires*, Argentina. Multiple courses.

Apr–Oct 2008 **Scientific Popularizer**, *Informatics dpt., University of Buenos Aires*, Argentina.

Publications

University of Lugano **Luis Mastrangelo**, Ph.D. Thesis, Advisors: Prof. M. Hauswirth and Prof. N. Nystrom. When and How Java Developers Give Up Static Type Safety. 2019

University of Lugano **Luis Mastrangelo**, M. Hauswirth and N. Nystrom. Casting about in the Dark: A Survey of Cast Usage in Java. 2019

University of Lugano **Luis Mastrangelo**, L. Ponzanelli, A. Mocci, M. Lanza, M. Hauswirth and N. Nystrom. Use at Your Own Risk: The Java Unsafe API in the Wild. 2015

University of Lugano **Luis Mastrangelo** and M. Hauswirth. JNIF: Java Native Instrumentation Framework. 2014

INRIA/University of Strasbourg A. Jimborean, **Luis Mastrangelo**, V. Loechner and P. Clauss. VMAD: an Advanced Dynamic Program Analysis & Instrumentation Framework. 2012

INRIA/University of Strasbourg	A. Jimborean, P. Clauss, B. Pradelle, **Luis Mastrangelo** and V. Loechner. Adapting the Polyhedral Model for Efficient Speculative Parallelization. 2012

Language Proficiency

Spanish	**Native speaker**
English	**Proficient User**: Level C2 under CEFR
Italian	**Proficient User**: Level C1 under CEFR
French	**Independent User**: Level B1 under CEFR